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CSPB 3112

24 April 2026

Data Pipeline: ART Linear Regression Model

Introduction:

The vision for this project was to implement an application with real-world implications both for patient education and provider-based patient classification. To accomplish this, the project focused on building a full data pipeline that integrates a front-end interface, an SQL database, Python-based data processing/analysis, and a central data repository. This project serves as a de facto capstone focusing on the data-driven pathway in the CSPB program. Goals for this project at its core were to perform a comprehensive data analysis lifecycle and link findings through different levels of programming to demonstrate an understanding of analytical reasoning and technical aspects of the key concepts in exploratory data analysis.

Background:

The project focuses on the exploratory data analysis lifecycle and the development of a foundational pipeline for which data can be accessed, processed, and presented. Additionally, this project is centered on using real-world data analyzed from a statistical standpoint to uncover insights that could have implications for patient care.

This project is important to my professional development because I intend to pursue a career in medical data analysis. The methods and tools used in this project align closely with industry practices allowing me to build relevant skills while applying data-driven approaches to healthcare problems.

Methodology, Materials, Methods:

To develop the knowledge and skills for this project, I selected specific concepts to use and implemented them. I used a variety of mediums to educate myself on individual topics, many of which I was already familiar with from my coursework. Specifically, the Geeks for Geeks website was instrumental in understanding and implementing many of the concepts I explored.

Additionally, I read the documentation of the various packages I used to both gain a better understanding of how they worked but to also explore some of their capacity. The combination of structured learning with independent study allowed me to develop my technical proficiency beyond the core curriculum.

Frameworks used:

Flask

Python: Numpy, Seaborn, Pandas, Matplotlib, SciKit Learn, SciPy

SQL: SQLite3

Results:

This project taught me several important lessons. Firstly, I reinforced the idea that progress isn't linear. While there were few setbacks throughout the process, I found reaching the benchmarks I set out to accomplish in my long-term vision somewhat unfulfilled. Secondly, I learned that there is something to be learned from statistical analysis even if metrics don't improve. Specifically, as I attempted to reduce my RMSE through various techniques it either didn't move at all or got worse. This result from the data allowed me to explore why it didn't change by analyzing different statistical values and was a testament as to why data analysis can be painstaking. Ultimately my findings were as expected and met my instinctual expectations. I didn't reach my goal of reducing the RMSE through data transformation or manipulation, but I did reach the overall goal of establishing the pipeline and performing statistical analysis.

Findings:

I used a consistent set of attributes for comparison through my statistical analysis: State == CA and Type == Donor frozen/embryos. The initial RMSE value was ~19. I tested two models: Model A and Model B. Model A was a simple additive model using the two features mentioned above one-hot encoded, and Model B used an interaction model where an additional third attribute was the combination of State and Type. Incorporating the third interactive term didn't produce any meaningful reduction in RMSE. To explore the relationship between the variables, I conducted a Chi-Square test of independence on my two original attributes. The Chi-Square test resulted in a statistically significant association between State and Type suggesting that the distribution of Type varies across different states and is not random.[1] Despite the statistical dependence between the two variables the interaction between the two didn't improve predictive performance. This finding shows that statistical dependence between variables doesn't imply interaction as an improvement variable in this regression model. Therefore, the attributes can be described as having an additive effect and their interaction doesn't contribute to additional variance.

[1] * X^2: 3810.36, p-value: <.001, dof: 86

Discussion/Reflection:

I did meet my goals for this project because it gave me exposure to the field I want to pursue and motivated me to move forward in a specific direction. I feel comfortable with what I accomplished and excited about where to go from here. Although it was just the tip of the iceberg, I thoroughly enjoyed the experience and look forward to pursuing mastery. A key factor that allowed me to meet my goals was to create a strong outline of what I wanted to accomplish and systematically check boxes along the way. I had a specific goal of creating a data pipeline and generating usable values as a basis for analysis. Once the pipeline was in place I was able to focus on the statistical analysis, which was the most engaging part of the project. Staying diligent and working consistently on the project helped me stay on track and ultimately result in a tangible deliverable. That said, I'm a little dissatisfied with the project results in that I wanted to perform a more comprehensive analysis of the data to have more distinct findings and more time than I wanted was spent in simply getting the pipeline to work properly. It seemed like every time I added something I broke the pipeline resulting in hours of debugging.

Conclusion:

As I reach the end of the program, I am proud of how far I've come. Previously, I had never written a single piece of code, and despite having performed some data analysis previously, I had never used any of the tools that I now understand. Currently, I'm focused on finishing this last semester strong with the same amount of effort that I have put in since the beginning. As for the future, I feel hopeful that this newfound knowledge is the beginning of a new professional journey. With an enhanced perspective I don't particularly believe in the 'AI is taking everyone's jobs' outlook because of the complexity that I now see in the field. Moving forward I plan to advance this project utilizing different machine learning methods as well as making the application more robust.

References Bibliography:

1. Supervised learning — scikit-learn 1.8.0 documentation
2. 1.1. Linear Models — scikit-learn 1.8.0 documentation
3. User Guide — pandas 3.0.2 documentation
4. chisquare — SciPy v1.17.0 Manual
5. Linear Regression in Machine learning - GeeksforGeeks